Exercise 32: create multiple db and users in mysql

1. Create db1, db2, db3 in mysql on the remote hosts
2. Create user1, user2, user3 in mysql on the remote hosts
3. Assign all privileges to the users (use nested loops)



Loop over a dictionary

Vars/main.yml

```
my_dict:
    Environment: production
    App: myapp
    Version: v1.0
```

Tasks/main.yml

```
- name: Loop over a dictionary
  debug:
    msg: "{{ item.key }} - {{ item.value }}"
  loop: "{{ my_dict | dict2items }}"
  tags: loop2
```




Loop over inventory

```
- name: loop over hosts inventory in my servers group
  debug:
    msg: "{{ item }}"
  loop: "{{ groups['myservers'] }}"
  tags: loop5

- name: loop over all the hosts in the current play
  debug:
    msg: "{{ item }}"
  loop: "{{ ansible_play_batch }}" or "{{ groups['all'] }}"
  tags: loop5
```




Pausing within a loop

گاهی اوقات بین حلقه‌ها نیاز هست مکث Pause داشته باشیم

Delay Loop control

✓ If you want to have a delay (in seconds) between the execution of each item in a task loop, you can use the pause directive with loop_control.

```
- name: loop over hosts inventory in my servers group
  debug:
    msg: "{{ item }}"
  loop: "{{ groups['all'] }}"
  loop_control:
    pause: 10
  tags: loop5
```




Tracking progress through a loop with index_var

✓ To keep track of where you are in a loop

مثل شمارنده حلقه هست
ولی با Loop control کار می‌کند

```
- name: Tracking the index
  debug:
    msg: "The {{ item }} index  is {{ my_index }}"
  loop:
    - first
    - second
    - third
  loop_control:
    index_var: my_index
```


Tracking progress through a loop with index_var

✓ To keep track of where you are in a loop

```
- name: Tracking the index
  debug:
    msg: "The {{ item }} index  is {{ my_index }}"
  loop:
    - first
    - second
    - third
  loop_control:
    index_var: my_index
```

Retrying a task until a condition is met

مثل while تا زمانی که شرط برقرار نشده است کار تو ادامه بده

```
- name: Retry a task until a certain condition is met
  shell: cat /home/test.txt
  register: result
  until: result.stdout.find("ansible") != -1
  retries: 3
  delay: 10
  tags: loop4
```




Import Playbook & Tasks

ما در playBooks می توانیم playBook دیگری هم فراخوانی کنیم

هر کدام از اینها جداگانه هستند

- Include playbooks inside a master playbook

Prj1.yml:

```
---
- hosts: test-hosts
  roles:
   - prj1
- import_playbook: prj2.yml
```

- Include tasks inside a master task

Prj1/tasks/main.yml:

```
- name: - - - - - task/main.yml - - - - -
  file: path=/home/test state=touch
owner=root group=root mode=0644
- import_tasks: task3.yml
```




Exercise 33: import playbooks and tasks

1. Create two playbooks in /home/ansible/provision named project1 and project2
2. Import project2 in project1 so it can execute after project1
3. In tasks directory create task2.yml and try to import it to the tasks/main.yml
4. Run the playbook!



Handlers

service ← موثرتر است stop - start - stop - مدیریت سرویس ها

- ✓ Trigger a task at the end of each block of tasks in a play, and will **only** be triggered **once** even if notified by multiple different tasks.
- ✓ Trigger a task if needed!

Tasks/main.yml:

```
- name: Handlers --- Stop Apache Tomcat
  template: src=tomcat_server.xml.j2 dest={{ Tomcat.Conf_Dir }}/server.xml
  notify:
    - restart tomcat *
```

Handlers/main.yml:

```
- name: restart tomcat*
  service: name=tomcat state=restarted
```

Start Tomcat once, only if the config file changed



Exercise 34: handlers

1. Try to change the apache configuration using template module
2. Add a handler to restart the apache service after the configuration
3. Change the configuration and add another step to reconfigure the apache
4. Both template steps should have notify to call the handlers
5. Run the playbook and check the status of the apache service



Ansible Vault

دیتاهای که در PlayBook قرار داره رمزنگاری می کنه
همینطور هم فایل ها و strings ها

- keeping sensitive data such as passwords or keys in **encrypted files**, rather than as plaintext in your playbooks or roles.
- **Any structured data** file can be encrypted like tasks, handlers, vars, bin, …
- An encrypted file as the 'src' argument of copy, template, unarchive and script modules automatically can be decrypted on the target.



Ansible Vault

- Create a new encrypted data file:

Ansible-vault create test.yml

ساخت یک فایل رمزنگاری شده

- Edit an encrypted file in place:

SHA256

Ansible-vault edit test.yml

فایل رمزنگاری شده است برای اینکه برگرده حالت قبل این رو با رمزش باز میکنیم

- Change password on a vault-encrypted file:

Ansible-vault rekey test.yml

- Encrypt an existing file:

خارج شدن

ansible-vault decrypt prj1.yml

عوض کردن رمز فایل

ansible-vault view prj1.yml

Ansible-vault encrypt test.yml

مشاهده فایل

رمزنگاری کردن فایل



فایلی که رمزنگاری کردیم می‌توان

مشاهده کردن

ادیت کردن

اجرا کردن

-- ask-vault-pass

با دادن رمز PlayBook اجرا شود

Exercise 35: ansible vault

1. Encrypt your current playbook with ansible vault
2. Try to run the encrypted playbook
3. Change the password of the encrypted playbook
4. Try to edit the encrypted playbook and run it again



Ansible Vault

✓ Single Encrypted Variable

ansible-vault encrypt_string admin@123 --ask-vault-pass

```
New Vault password:
Confirm New Vault password:
!vault |
      $ANSIBLE_VAULT;1.1;AES256
      39373934366439663565613436643633383632653837333037633330663664363065643834636435
      63623566616462666633316532316332373335336634303 30a396366323336616235333936613366
      36373032316135353665346633313964393037636532346161663934653432663731326536653664
      36343531373766666340a6130653230383266373839363236346439316263303738396338323235 61
      6134
Encryption successful
```

✓ Replace the password in the yml file with the generated encrypted string

ansible-playbook -i inventory/AnisaHosts myproject.yml --ask-vault-pass



Exercise 36: ansible vault

1. Add a user named user1 and set a password for it
2. Encrypt the password string with ansible vault (do not encrypt the whole file)
3. Use no_log: true in the step
4. Run the encrypted playbook to add the user



Ansible Vault

- **Vault IDs** help in encrypting different files with different passwords to be referenced inside a playbook.

✓ Create multiple password files:

اگر ما فایل های PlayBook مون رو رمزگذاری های مختلف کرده باشیم

نمیشه از رفرنسیان استفاده کرد باز Vault ID استفاده کرد

```
# echo "123" > ~/.vault-pass.common
# echo "456" > ~/.vault-pass.staging
# echo "789" >~/.vault-pass.production
```

✓ Add vault_identity_list to ansible.cfg:

اسم گذاشتن برای فایل های Pass ما

```
# cat /etc/ansible/ansible.cfg
    [defaults]
    vault_identity_list = common@~/.vault-pass.common, staging@~/.vault-
    pass.staging, production@~/.vault-pass.production
```




Ansible Vault

✓ Encrypt files via vault-id:

```
# ansible-vault encrypt --encrypt-vault-id common test1.yml
# ansible-vault encrypt --encrypt-vault-id staging test2.yml
```

To run the PlayBook:

```
# ansible-playbook –i inventory/AnisaHosts myproject.yml
```

Note: If the first password won't open the vault, it will move on to the next one, until one of them works



Ansible Vault

- ✓ If you want to be prompted for password to decrypt the vault string/file:
  - ✓ comment out vault_identity_list key in ansible.cfg
  - ✓ execute the playbook with --vault-id id@prompt

ansible-playbook -i inventory/AnisaHosts --vault-id common@prompt --vault-id staging@prompt myproject.yml

Vault password (common):
Vault password (staging):



Exercise 37: Setup a webserver

1. Install java-1.8.0-openjdk-devel
2. Download Apache-Tomcat
   1. Url: https://ftp.wayne.edu/apache/tomcat/tomcat-9/v9.0.41/bin/apache-tomcat-9.0.41.tar.gz
3. Install and configure Tomcat
   1. Extract the apache-tomcat-9.0.41.tar.gz to /opt and make soft link to /opt/tomcat
   2. Configure Tomcat to listen port 80
   3. Create Tomcat service
      1. /etc/systemd/system/tomcat.service
4. Deploy the sample war file
5. Configure the firewall to allow port 80
6. Start Apache-Tomcat Service

